

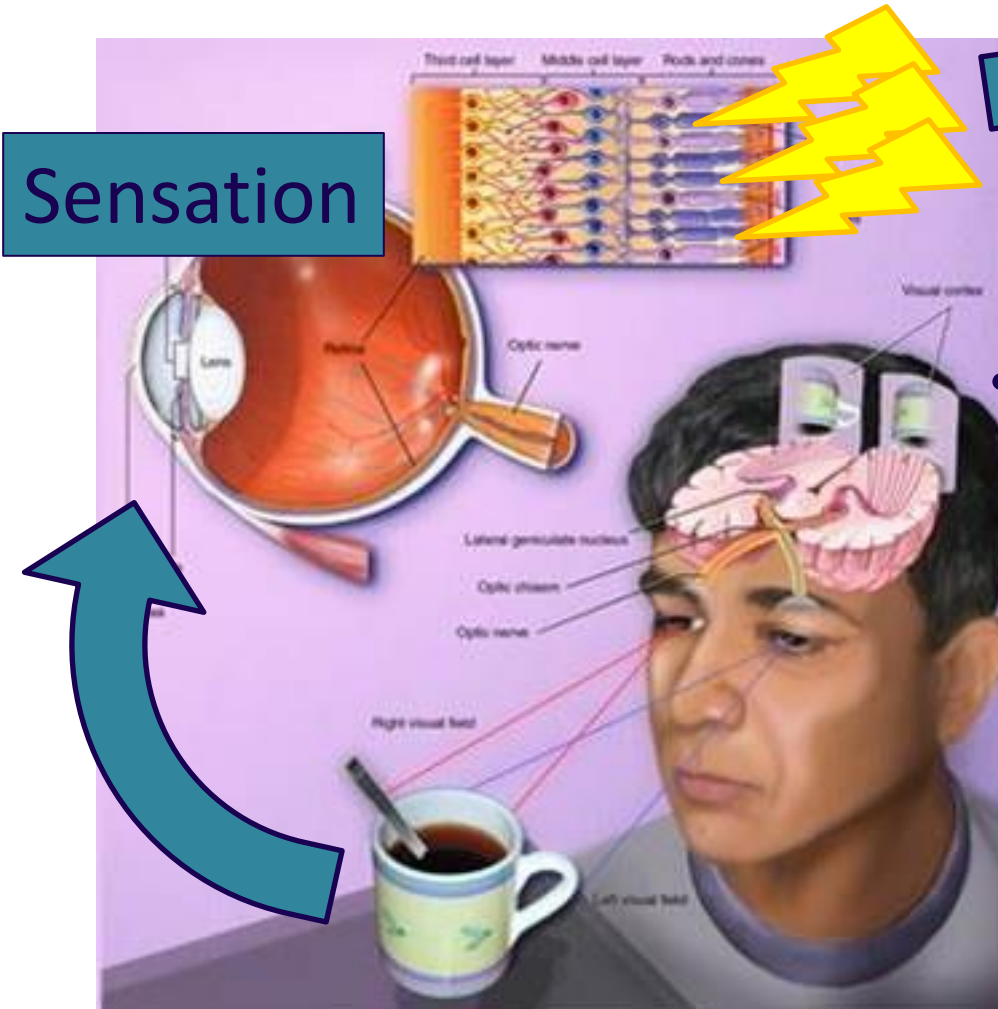
# Size perception

Prof Anna Grubert

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# So far, we have talked about...

Sensation



Knowledge

Perception

Colour

Depth

Size

# How do such *relative size illusions* work???

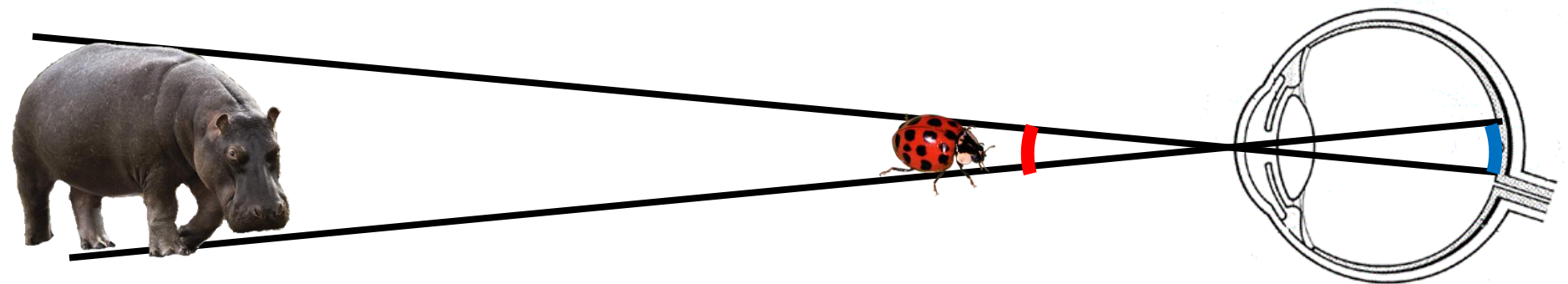


# Size perception

- Size of retinal object representations
  - Measured in degree of visual angle.

The angle of an object relative to the observer's eye.

The size of the retinal object representation.



Two differently sized objects can have **the same visual angle** when the larger one is further away.

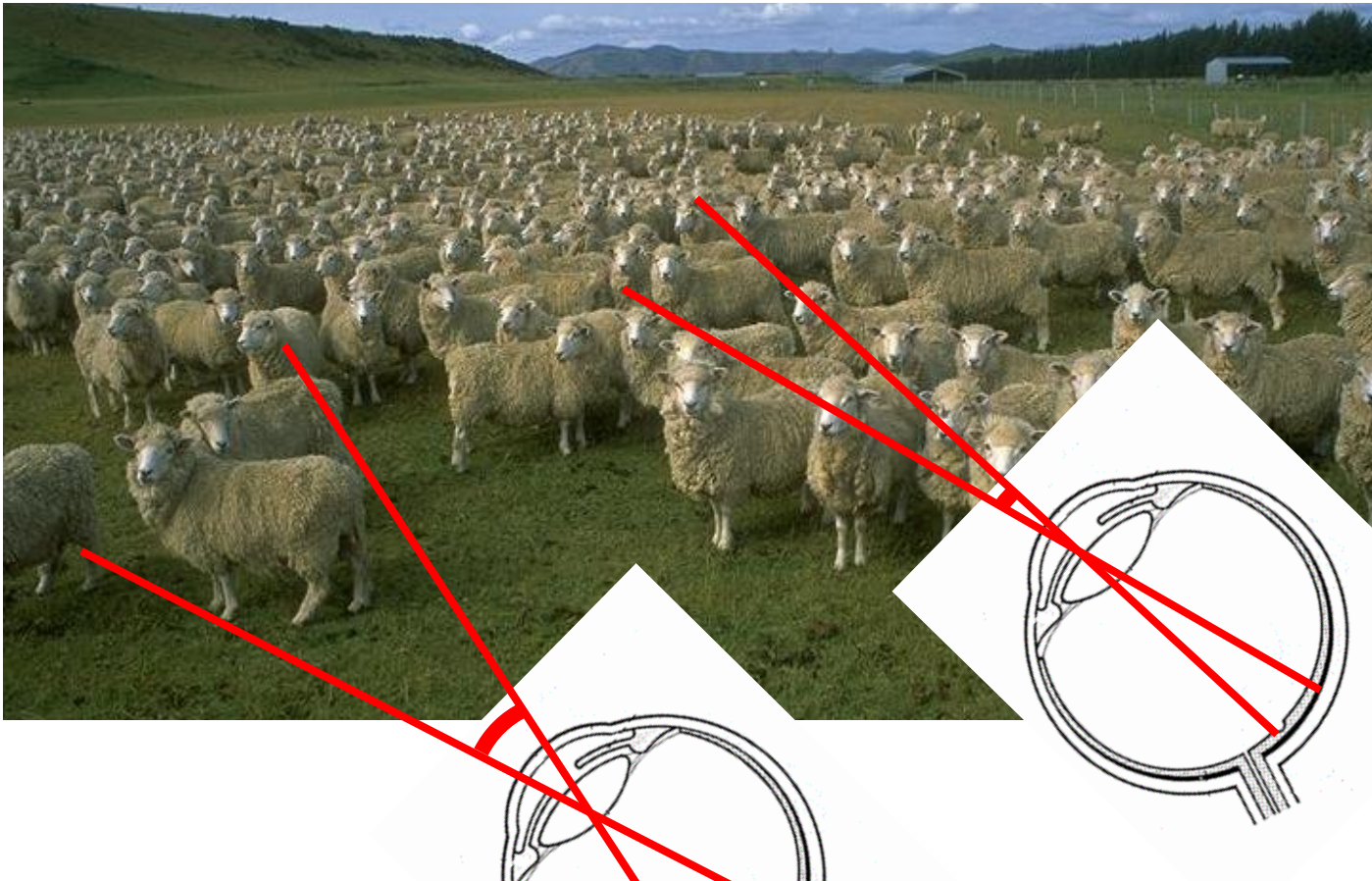




# Size perception

- Size constancy

**Perception** of an object's size is relatively **constant** even when viewed from different distances (producing different visual angles).



# Size perception

- Calculating size constancy
  - **Size-distance scaling equation**

Far away sheep: 2 = K(1x2)

Near sheep: 2 = K(2x1)

$$S = K(R \times D)$$

S  
Perceived size

R  
Size of the  
retinal image

D  
Perceived  
distance

If a sheep walks towards you, the size of the sheep's image on your retina (R) gets larger, but your perception of the sheep's distance (D) gets smaller.

The perceived size (S) of the sheep remains the same.



# Size perception

Size-distance scaling equation explains relative size illusions

$$S = K(R \times D)$$

Perceived size      Retinal image size      Perceived distance

Dino:  $2 = K(2 \times 1)$

People:  $1 = K(1 \times 1)$



The retinal image size is larger for the dino than the people.

Lack of pictorial depth cues:  
We assume that the distance of the dino and the people is the same.

Hence

The size of the dino is perceived to be bigger than the size of the people.

# Size perception

- Size-distance scaling equation explains the Ponzo illusion.

Rabbit 1:  $2 = K(1 \times 2)$

Rabbit 2:  $1 = K(1 \times 1)$

$$S = K(R \times D)$$

Perceived size    Retinal image size    Perceived distance



Rabbit 1

Rabbit 2



R: Rabbit 1 = Rabbit 2.

D: Rabbit 1 is perceived to be further away (pictorial depth cue).



Hence

S: Rabbit 1 is perceived to be larger.



# Size perception

- Size-distance scaling equation explains the moon illusion.



Moon 2



Moon 1

$$S = K(R \times D)$$

Perceived size    Retinal image size    Perceived distance

R: Moon 1 = Moon 2.

D: Moon 1 is perceived to be further away (pictorial depth cues).

Hence

S: Moon 1 is perceived to be larger.

# Size perception

- Same explanation for the Ponzo and moon illusion.

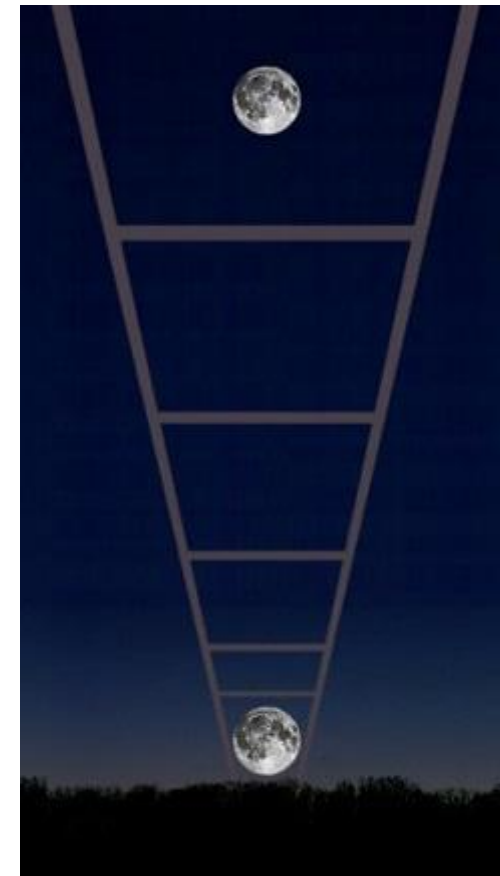
Rabbit 1:  $2 = K(1 \times 2)$

Rabbit 2:  $1 = K(1 \times 1)$



Moon 1:  $2 = K(1 \times 2)$

Moon 2:  $1 = K(1 \times 1)$



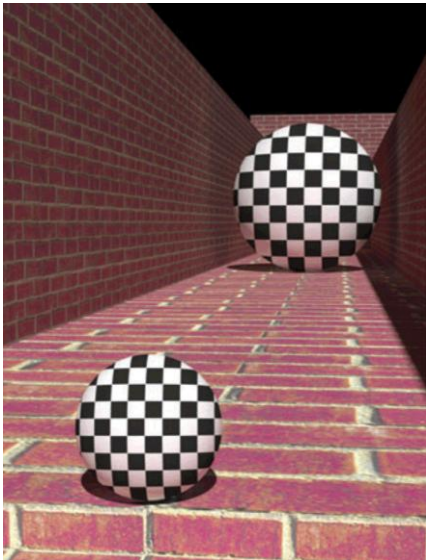
Moon 2

Moon 1

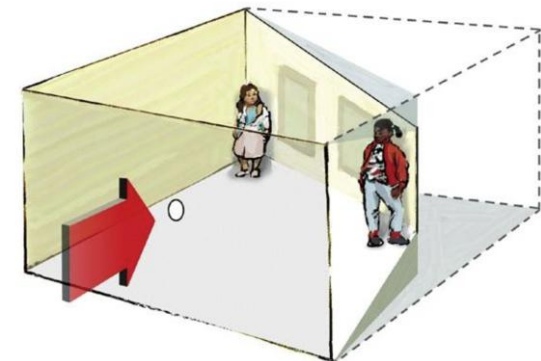
# Try it for yourself...

- Fill in the size-distance scaling equation for these illusions:

Hint: the 2 balls are equal in size



Hint: the room in which the photo was taken looks like this:



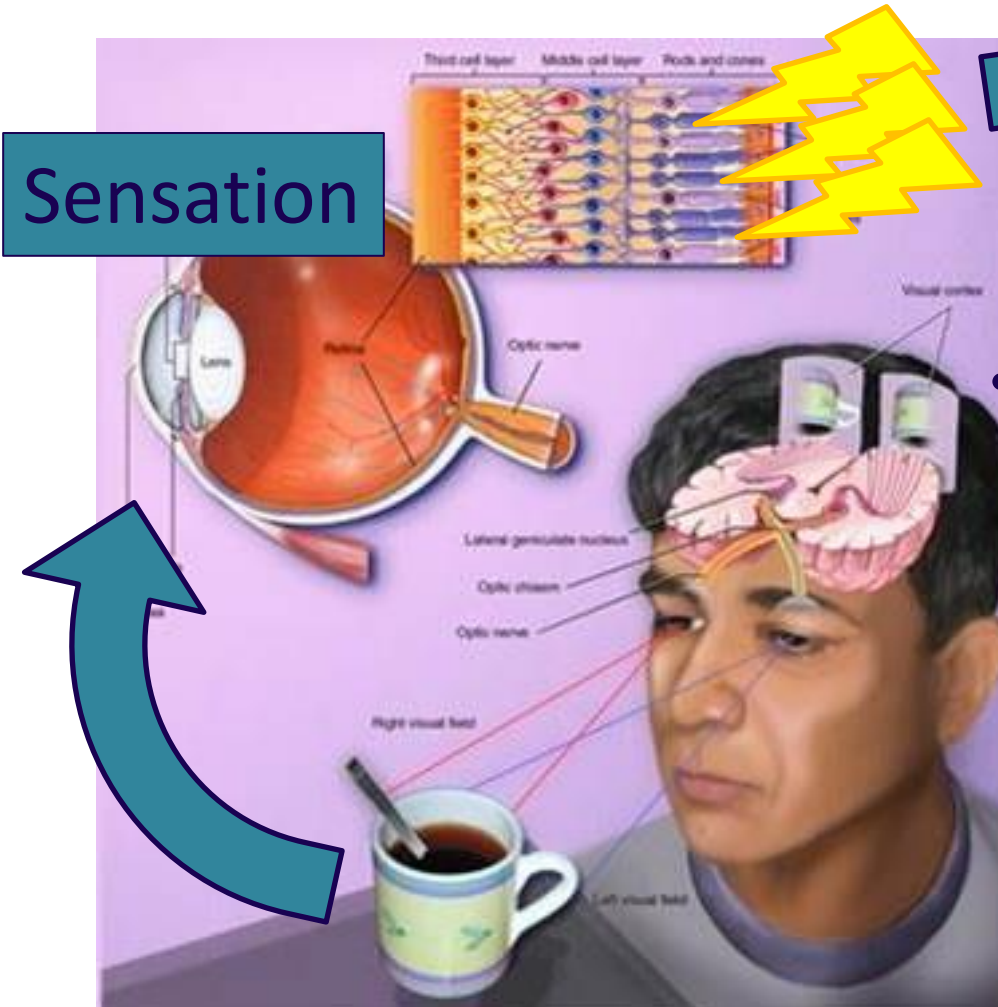
# Motion perception

Dr Anna Grubert  
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# We also want to talk about...

Sensation



Knowledge

Perception

Colour

Depth & size

Motion

# What does motion perception do for us?

- Continuous environmental update



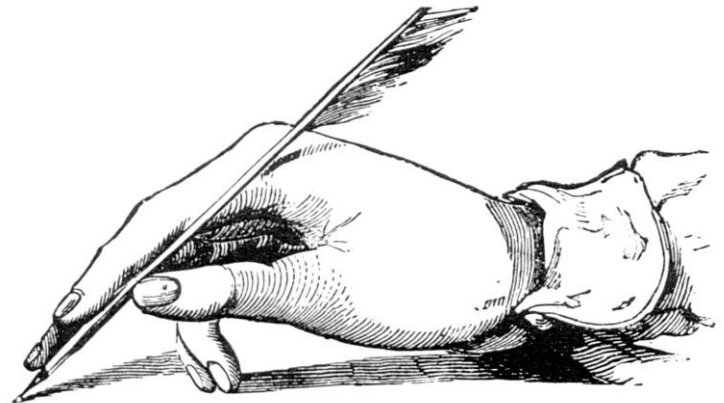


do for us?



# What does motion perception do for us?

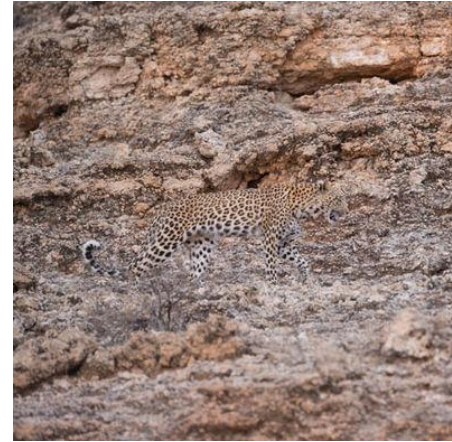
- Continuous environmental **update**
  - Try **writing**...
    - ...but you're only allowed to move your pen when your eyes are closed.





# What does motion perception do for us?

- Grouping and segmentation





# What does motion perception do for us?

- Grouping and segmentation



# Ways to perceive motion

- **Real motion**
  - A stimulus that really moves.  
Look at me now...



# Ways to perceive motion

- **Illusory motion:** Apparent motion
  - Perceived motion in a static stimulus.



# Ways to perceive motion

- **Illusory motion:** Apparent motion
  - Perceived motion in a static stimulus.



# Ways to perceive motion

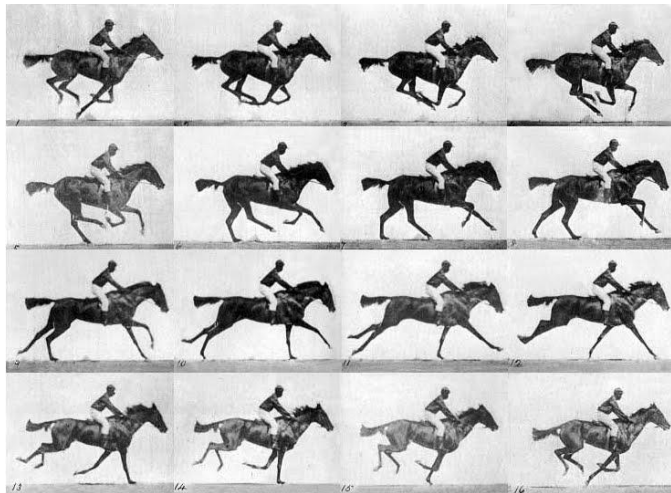
- **Illusory motion:** Apparent motion
  - Perceived motion in a static stimulus.





# Ways to perceive motion

- **Illusory motion:** Apparent motion
  - Illusion of motion for successively presented stimuli.  
e.g., film



19<sup>th</sup> century TV



# Ways to perceive motion

- **Illusory motion:** Induced motion
  - A moving **frame of reference** (e.g., clouds) can produce the illusion that a **stationary object** (e.g., moon) moves in the opposite direction.

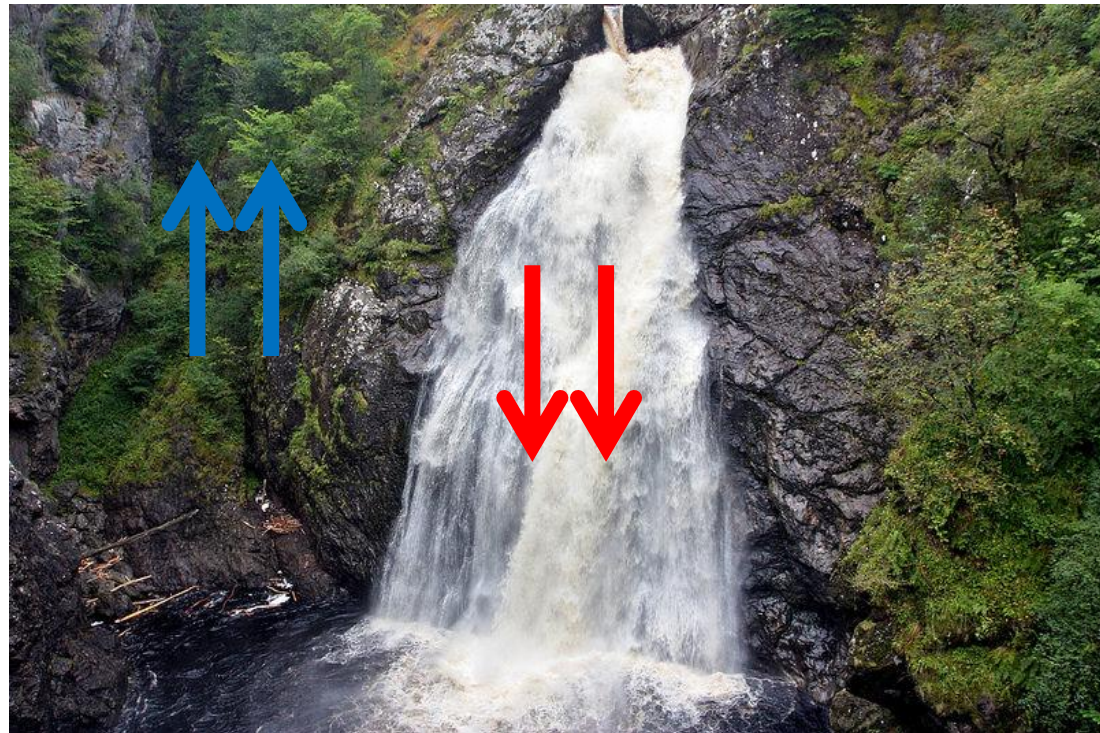


# Ways to perceive motion

- **Illusory motion:** Motion aftereffects
  - After observing (adapting to) **motion in one direction** (e.g., waterfall), **stationary objects** (e.g., trees) appear to move in the opposite direction.

If you want to know more about **neuronal adaptation** read this review (Anstis, Verstraten, & Mather, 1998, TICS):

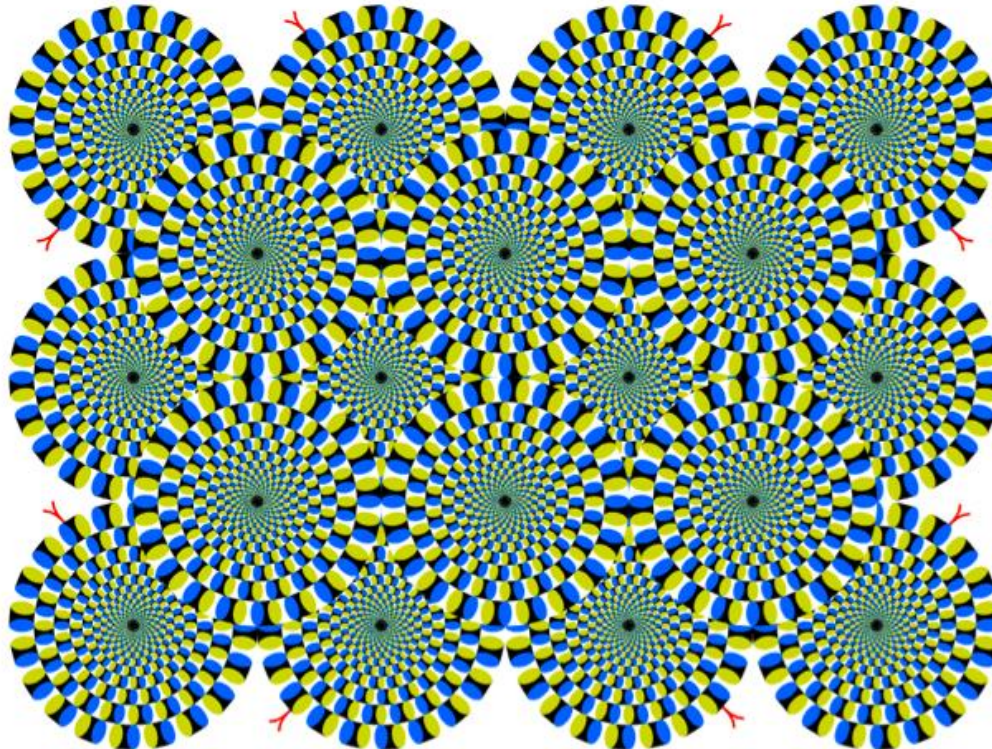
<https://quote.ucsd.edu/anstislab/files/2012/12/1998-the-motion-aftereffect.pdf>





# Ways to perceive motion

- **Illusory motion:** Visual illusions
  - e.g., Peripheral drift illusion.



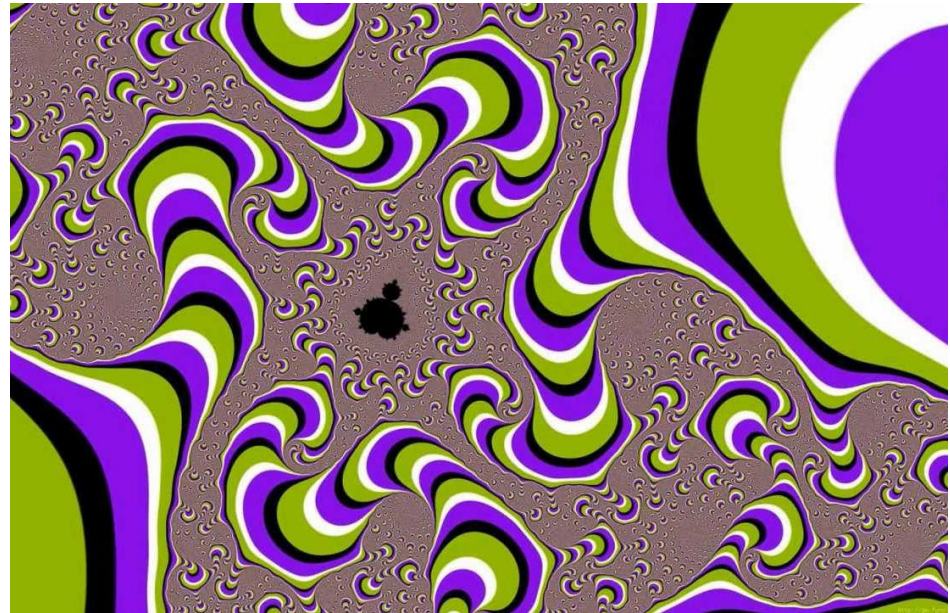
More of that:  
<http://www.rits.umei.ac.jp/~akitakoka/index-e.html>

# Ways to perceive motion

- However!
  - Only real motion produces **retinal motion**.  
Retinal motion = **successive receptor stimulation** by a moving stimulus in the visual field.

e.g., motion illusion.

**No retinal motion,**  
happens when the  
very same receptors  
are stimulated.

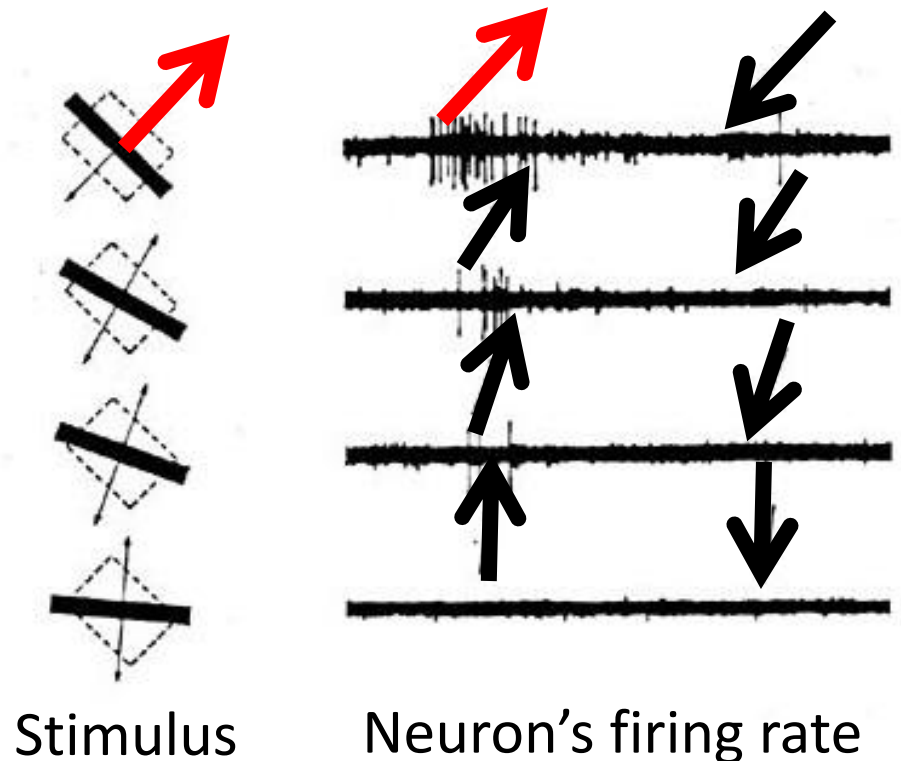
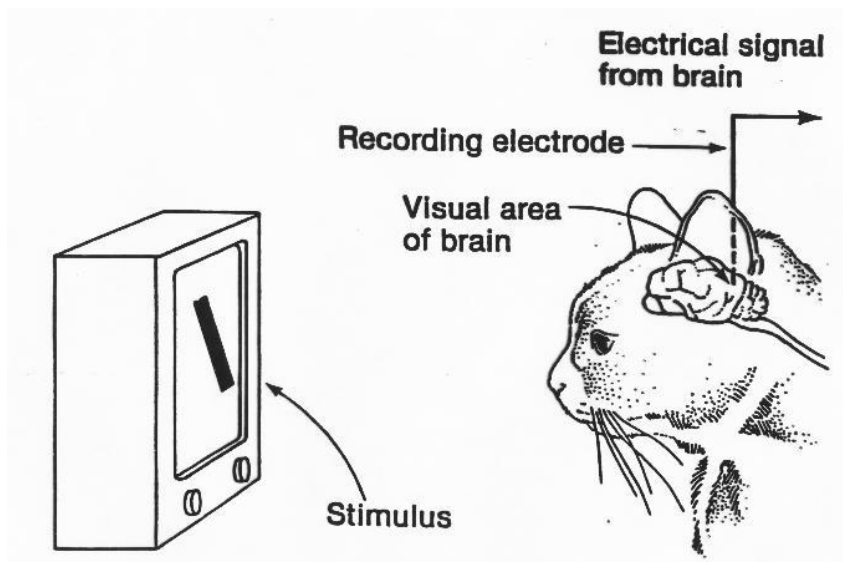


# Perception of real motion

- Physiological evidence

- Motion-direction selective** neurons in V1

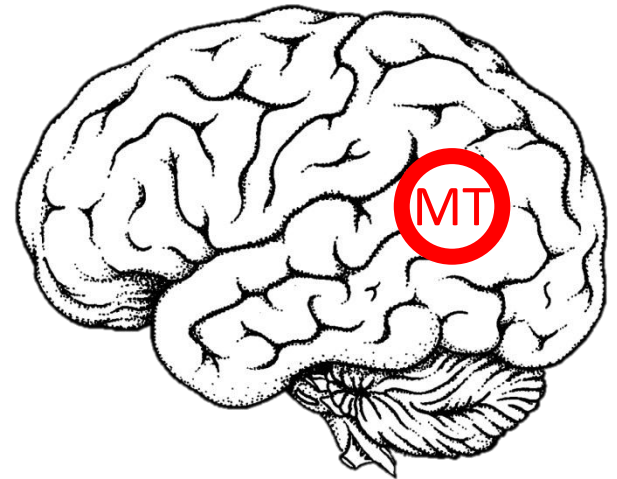
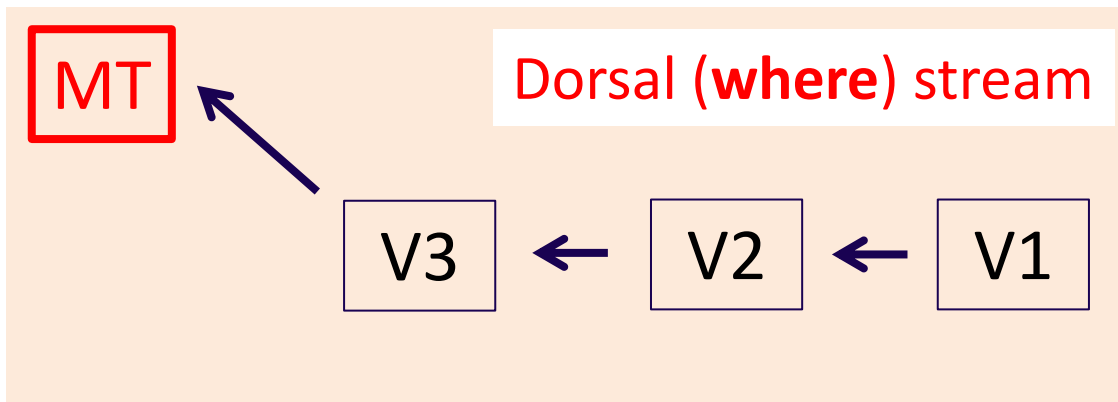
Hubel & Wiesel, 1959





# Perception of real motion

- Physiological evidence



**Middle temporal area (V5):**

Almost all neurons are direction-selective  
(to process **where** an object is).

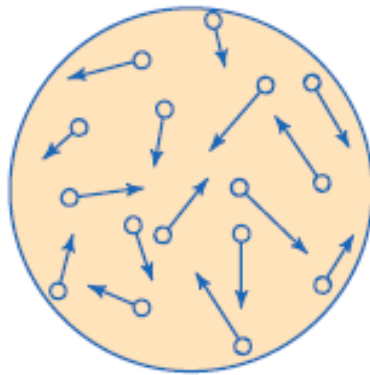
# Perception of real motion

- Physiological evidence

- Newsome et al., 1995

Monkey experiment with **moving dot displays**.

Measured an MT neuron's firing rate (**single-cell recording**) depending on motion direction coherence.



# Perception of real motion

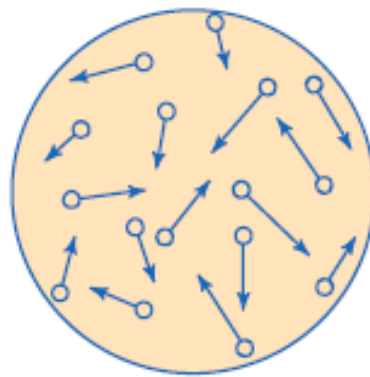
- Physiological evidence

- Newsome et al., 1995

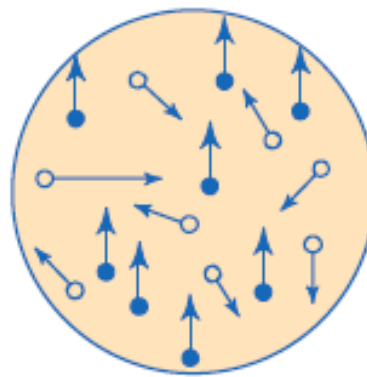
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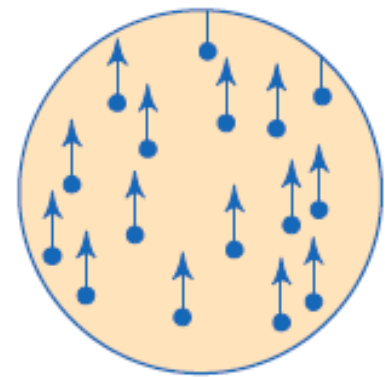
**Coherence**  
of the dots'  
motion  
direction.



Random  
motion



50% same  
direction



100% same  
direction



# Perception of real motion

- Physiological evidence

- Newsome et al., 1995

With **increasing** motion direction **coherence**, ...

...monkeys judged the direction of motion **more accurately**.

...the MT neuron fired **more rapidly**.

→ 25 out of 200 dots...

With **12.8% coherence**, monkeys judged motion direction accurately in almost every trial.

**Conclusion:** MT neurons are highly specialised to detect motion direction.

Why is this so fantastic?

# Perception of real motion

- Physiological evidence
  - A single neuron's receptive field only covers a part of the visual field.

**Receptive field** =  
Part of the visual  
field in which a  
stimulus can  
modify a neuron's  
firing rate.



# Perception of real motion

- Physiological evidence
  - A single neuron's receptive field only covers a part of the visual field.

No neuron responds to the whole visual field.  
This is a **problem** for motion perception...

Schematic  
illustration of  
neuron A, B and  
C's receptive  
fields.



# Perception of real motion

- Physiological evidence

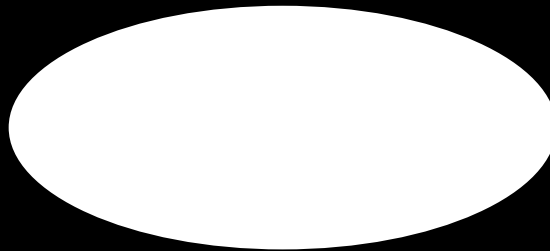
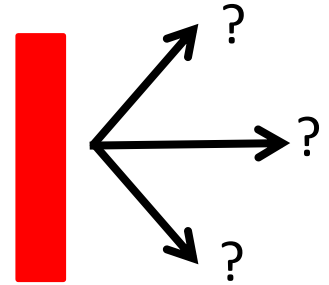
- The aperture problem

The **direction** of a moving stimulus through an aperture (such as a receptive field) **is ambiguous**.



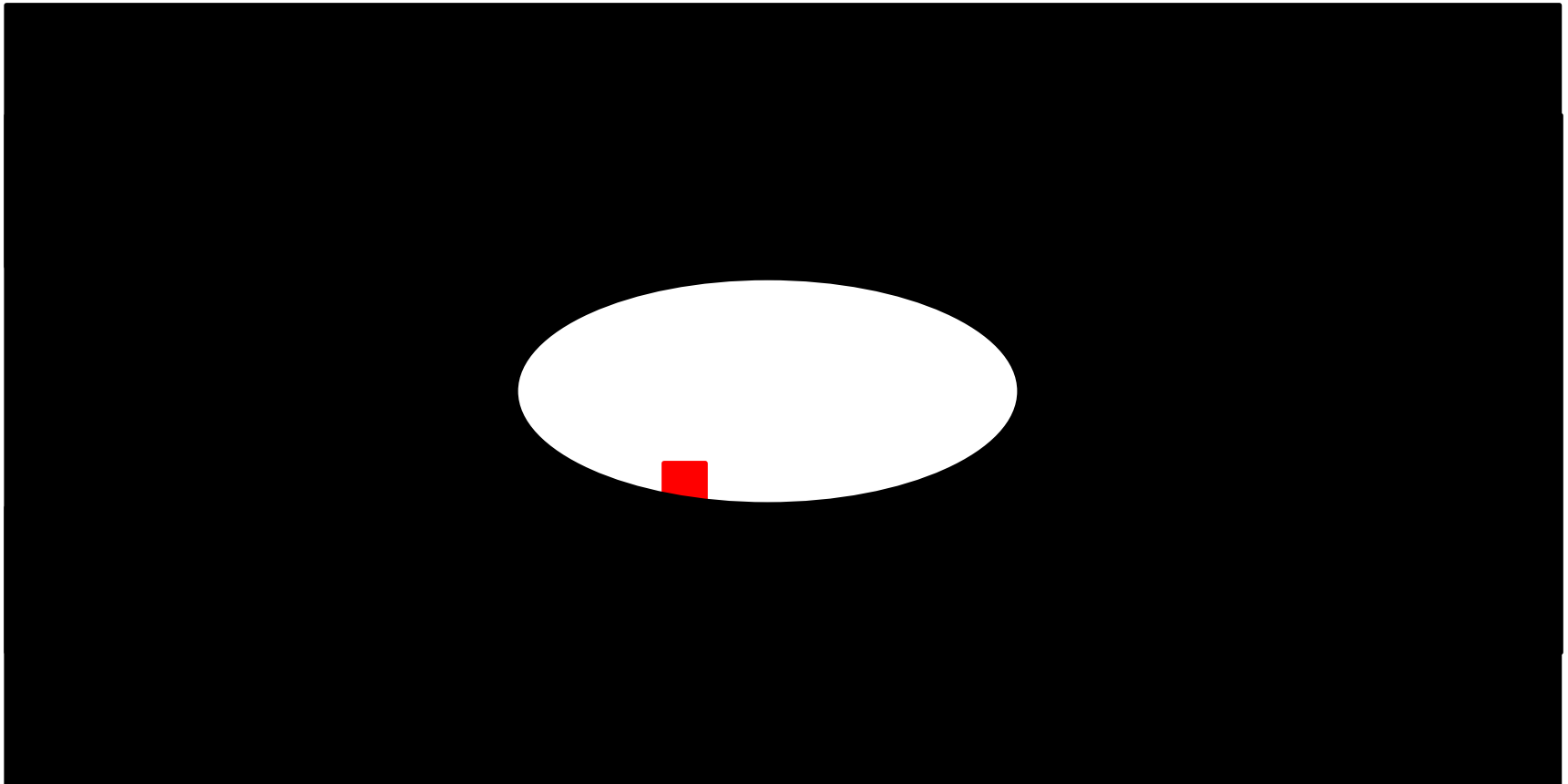
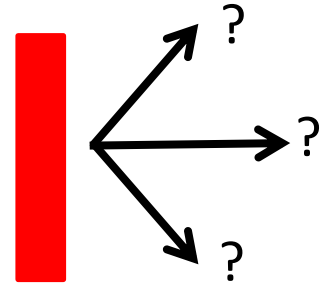
# Perception of real motion

In which direction does the bar move?



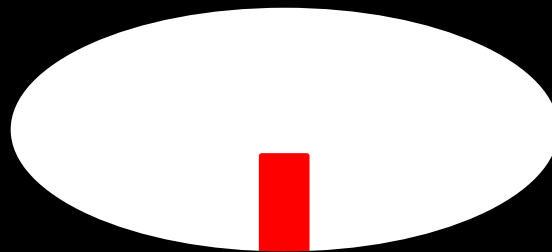
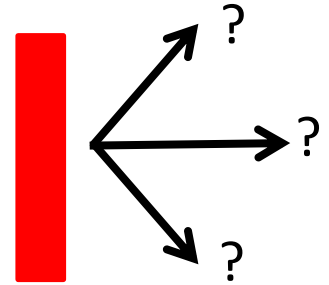
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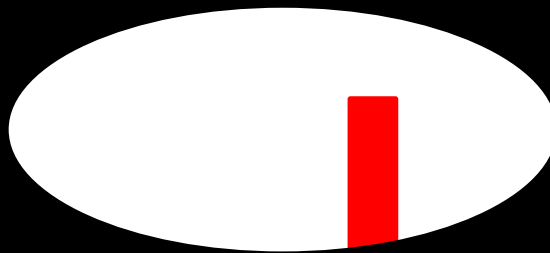
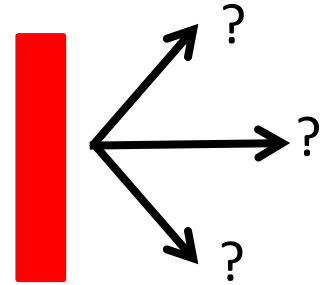
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# Perception of real motion

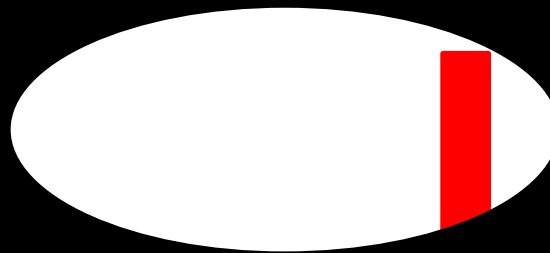
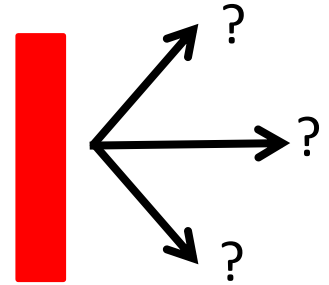
In which direction does the bar move?





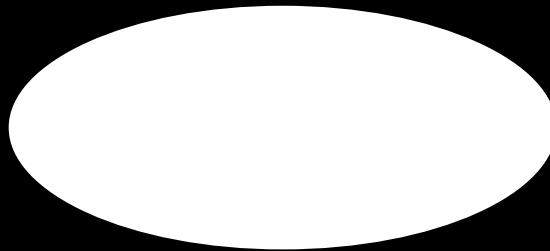
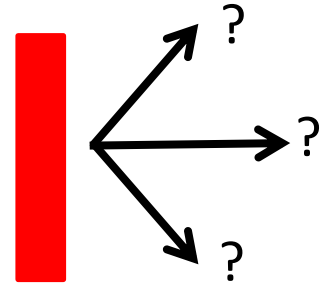
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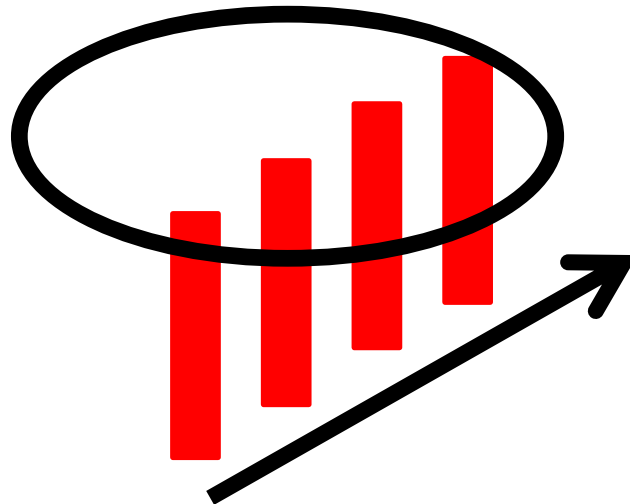
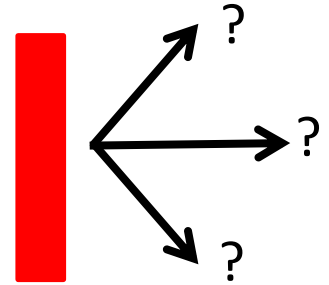
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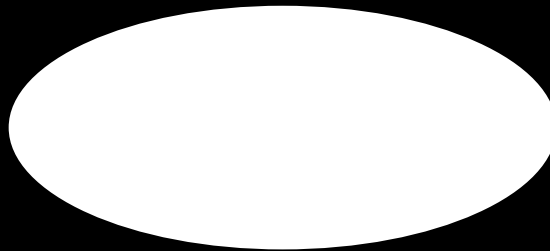
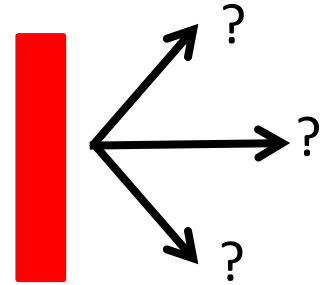
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In which direction does the bar move?



# Perception of real motion

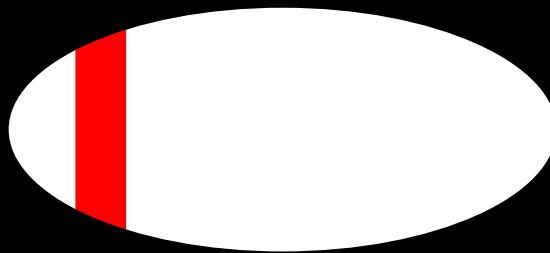
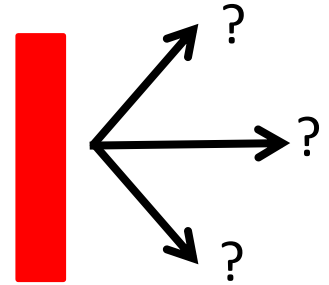
In which direction does the bar move?





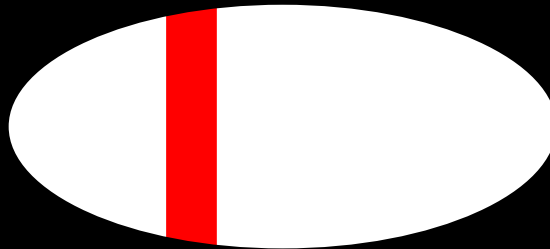
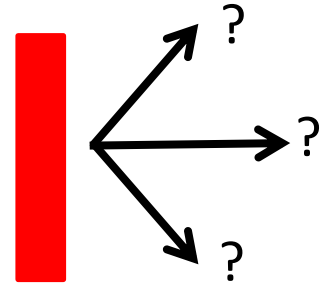
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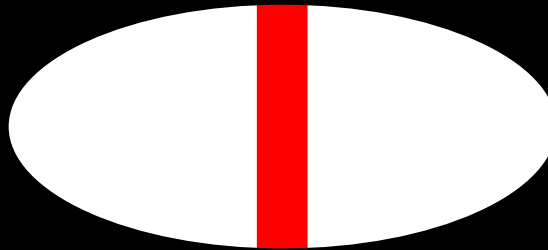
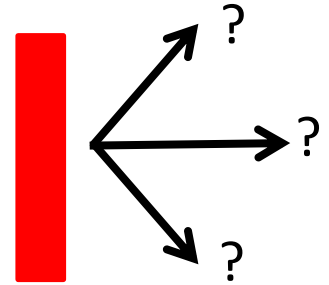
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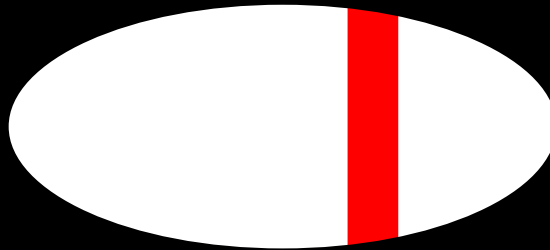
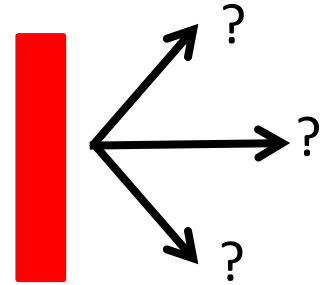
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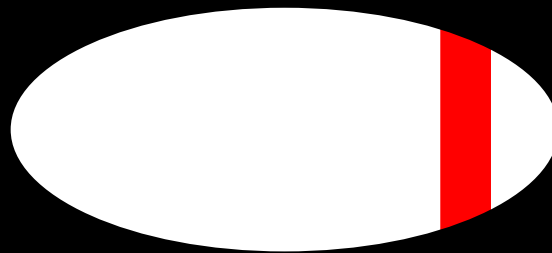
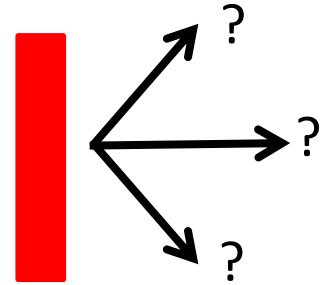
# Perception of real motion

In which direction does the bar move?



# Perception of real motion

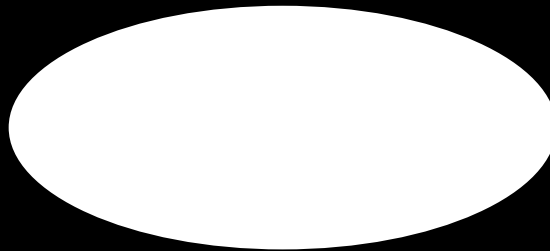
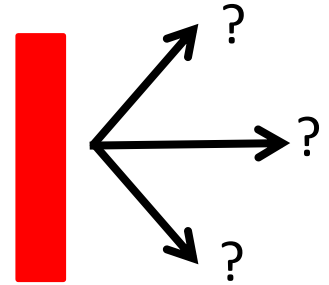
In which direction does the bar move?





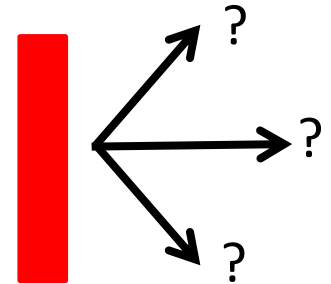
# Perception of real motion

In which direction does the bar move?

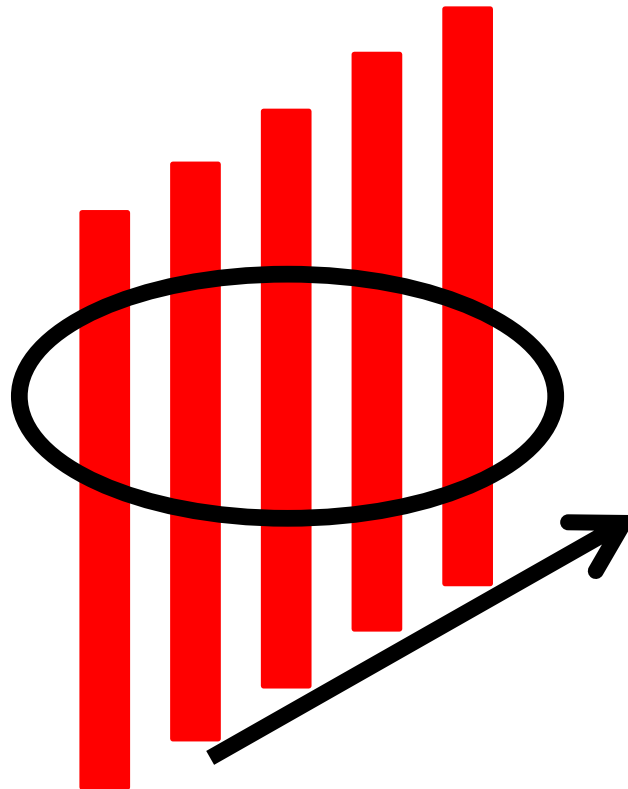


# Perception of real motion

In which direction does the bar move?



The motion direction is unclear when the stimulus is larger than the receptive field.



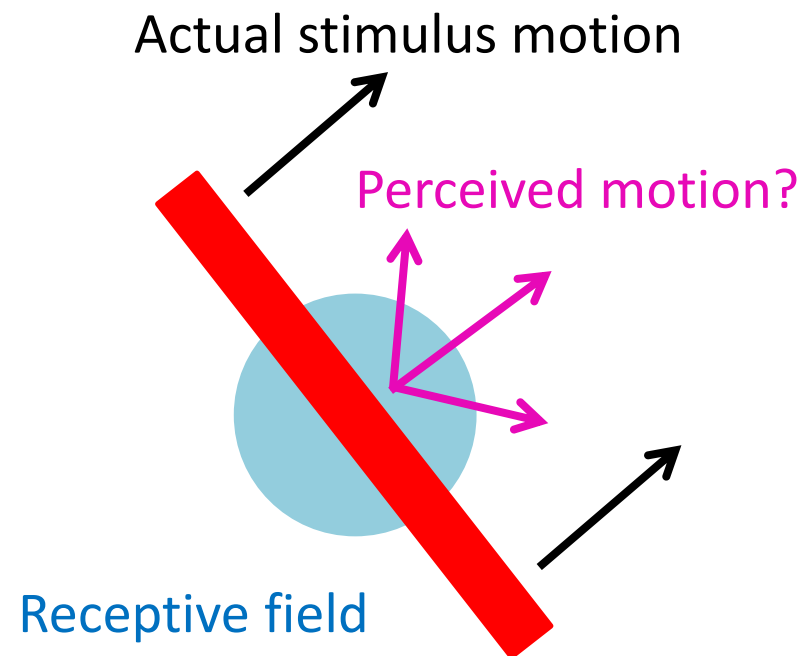
# Perception of real motion

- Physiological evidence

- The aperture problem

The **direction** of a moving stimulus through an aperture (such as a receptive field) is **ambiguous**.

The response of an individual direction-selective neuron does not provide sufficient information to indicate the direction of the movement.



# Perception of real motion

- Physiological evidence

- The aperture problem

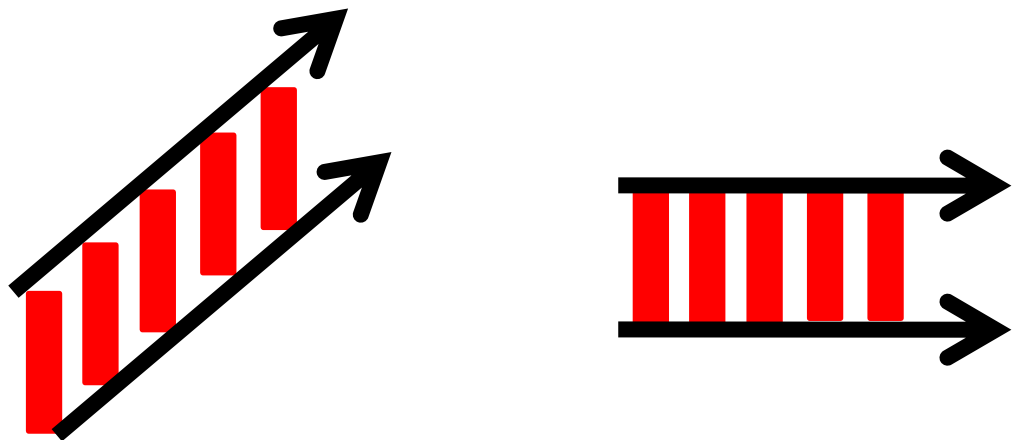
- Solutions

Neuronal convergence

1. **Pooling** responses of multiple neurons.

2. **End-stopped cells** signal the end of a stimulus (where the direction is unambiguous).

Object segmentation

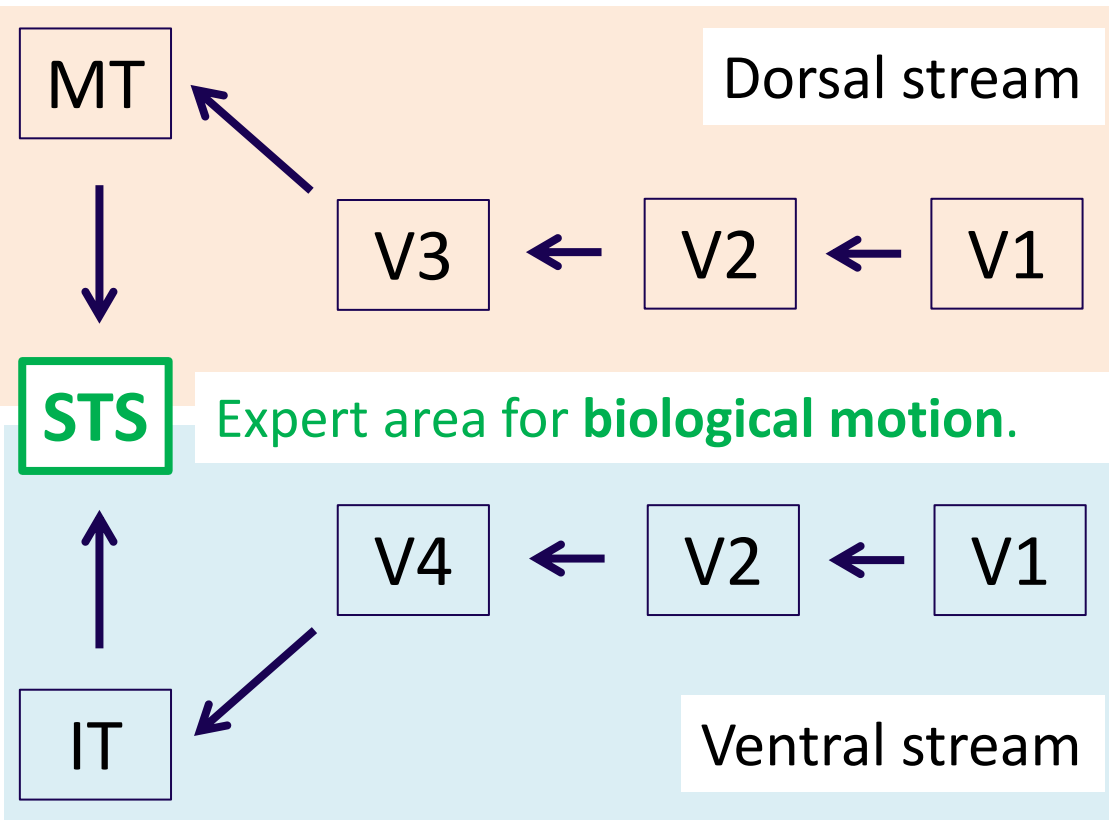


# Perception of real motion

- Physiological evidence

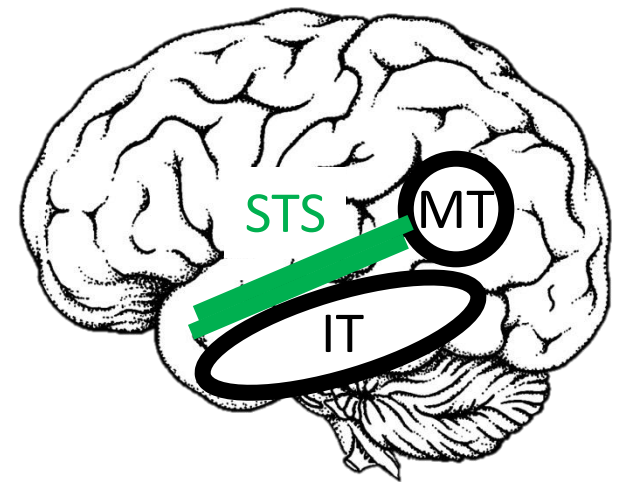
**STS**: Receives projections from MT and IT; combines input about object form and motion direction.

Motion



Form

STS = superior temporal sulcus



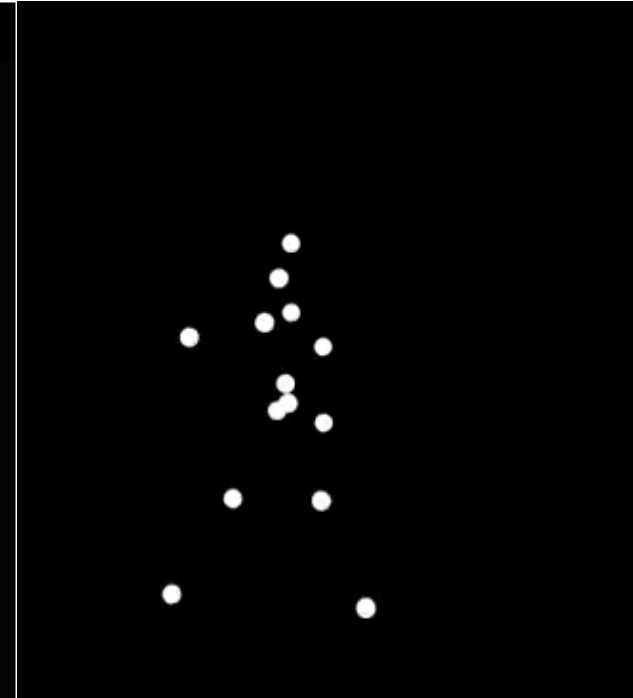
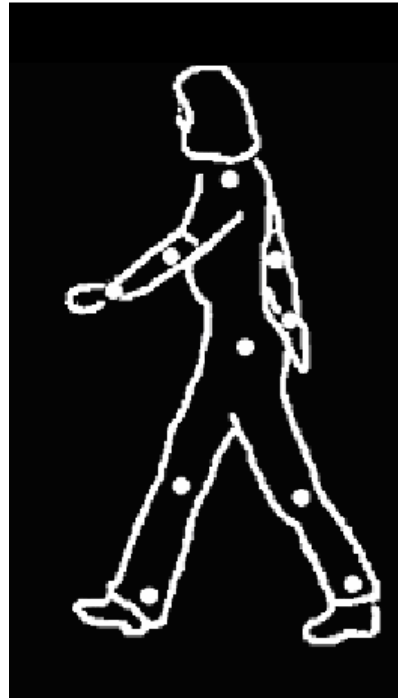
MT = Middle temporal area (V5)

IT = Inferotemporal cortex



# Perception of real motion

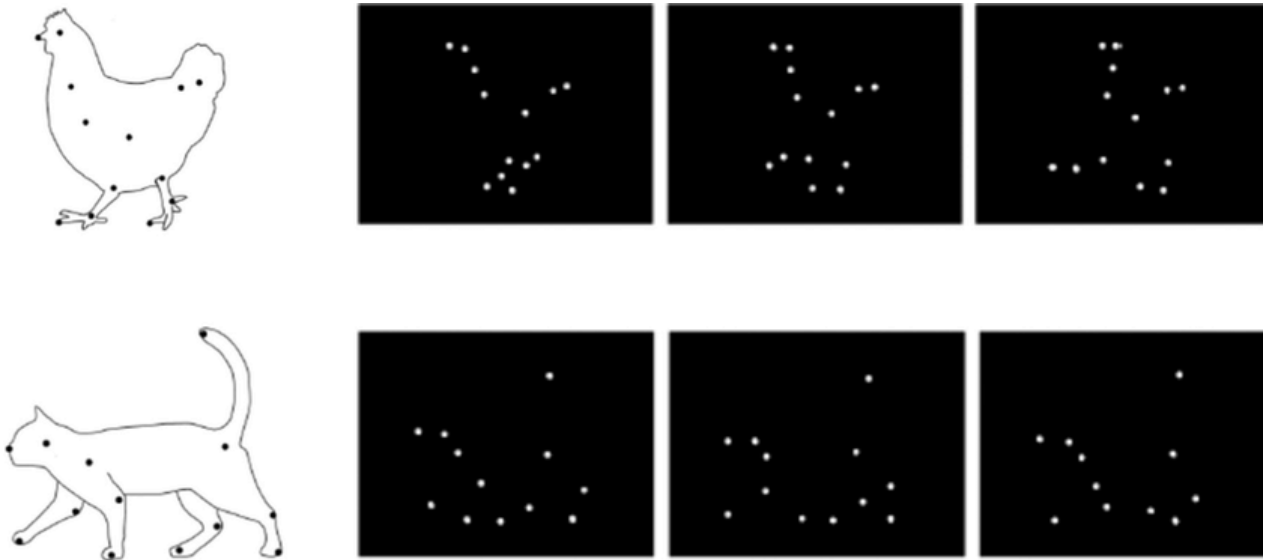
- Biological motion
  - **Self-produced motion** of a biological being.  
Real motion that produces retinal motion.
  - We are **experts** in perceiving and recognising biological motion.  
e.g., Point-light walkers.



# Perception of real motion

- Biological motion

e.g., Point-light chicken and cat



# Perception of real motion

- Only real motion produces **retinal motion**.
  - Retinal motion = **successive receptor stimulation**.

# Perception of real motion

If motion is perceived by retinal motion, why don't we perceive motion when we move our eyes from A to B?

We should, because this **stimulates receptors successively...**

**A**



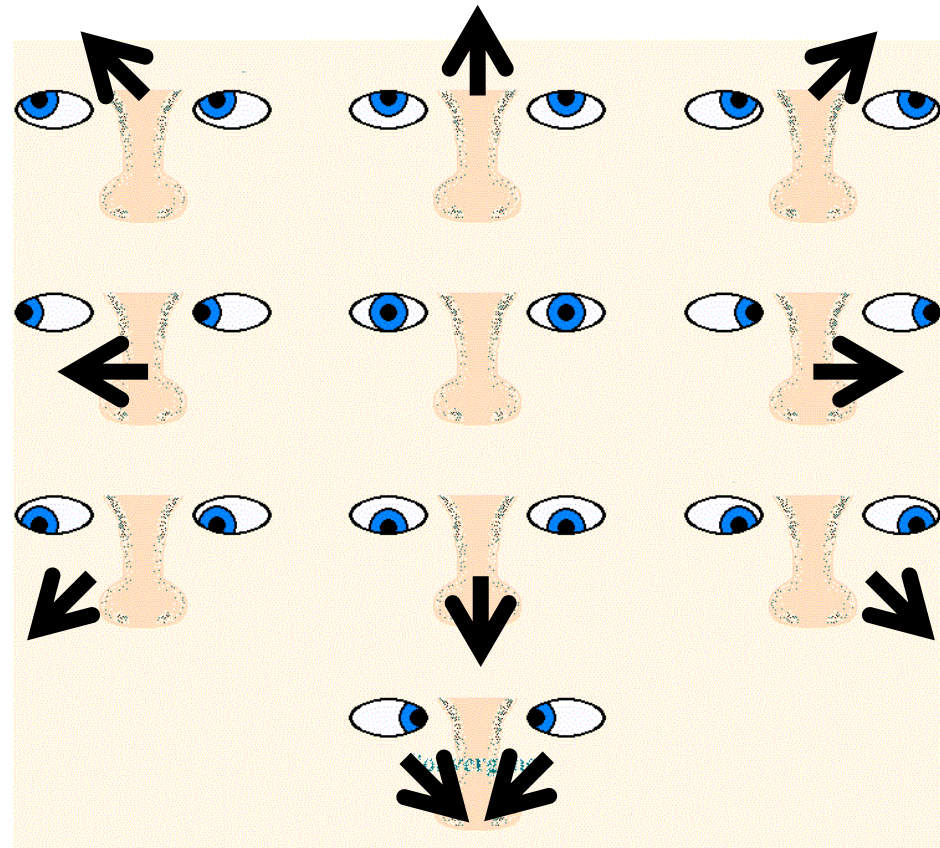
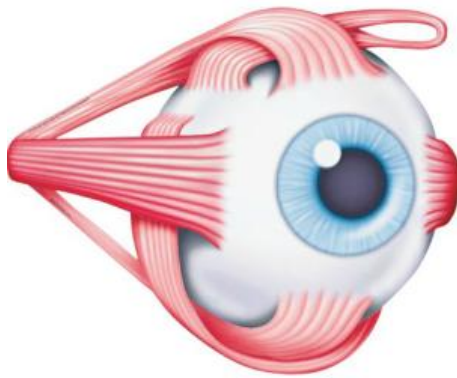
**B**

# Perception of real motion

- Corollary Discharge Theory
  - Motion perception depends on retinal motion **AND** eye movements.

# How do eye movements work?

- Eye muscles
  - 4 at the top, bottom, left, and right of each eye, 2 wrapped around it.



Different types of eye movements:

<https://www.youtube.com/watch?v=6GliSCGkpZ4&list=PL6hslGPUQLHbDBB90owlz8RdeGGNorHD7PS>



# Perception of real motion

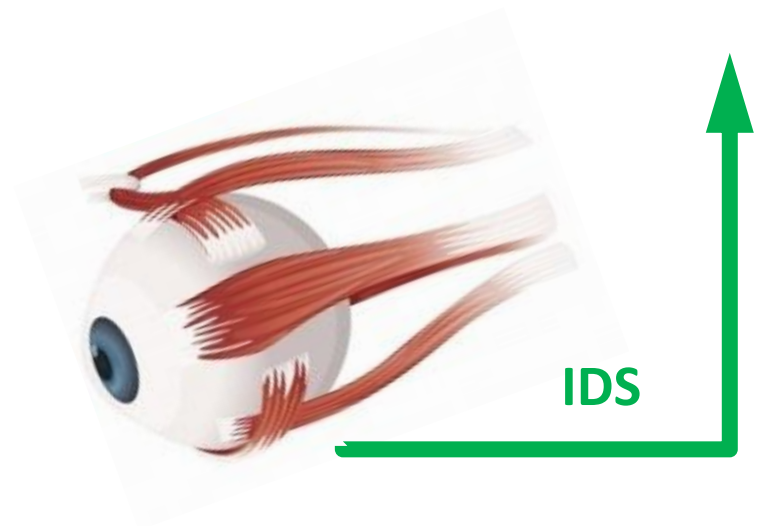
- Corollary Discharge Theory
  - Motion perception depends on retinal motion **AND** eye movements.
  - **3 relevant signals:**
    1. Image displacement signal
    2. Motor signal
    3. Corollary discharge signal

Comparison of these signals reveals whether or not **real motion** has occurred.

# Perception of real motion

## Image displacement signal (IDS)

An image moves across the retina and successively stimulates the receptors (retinal motion).



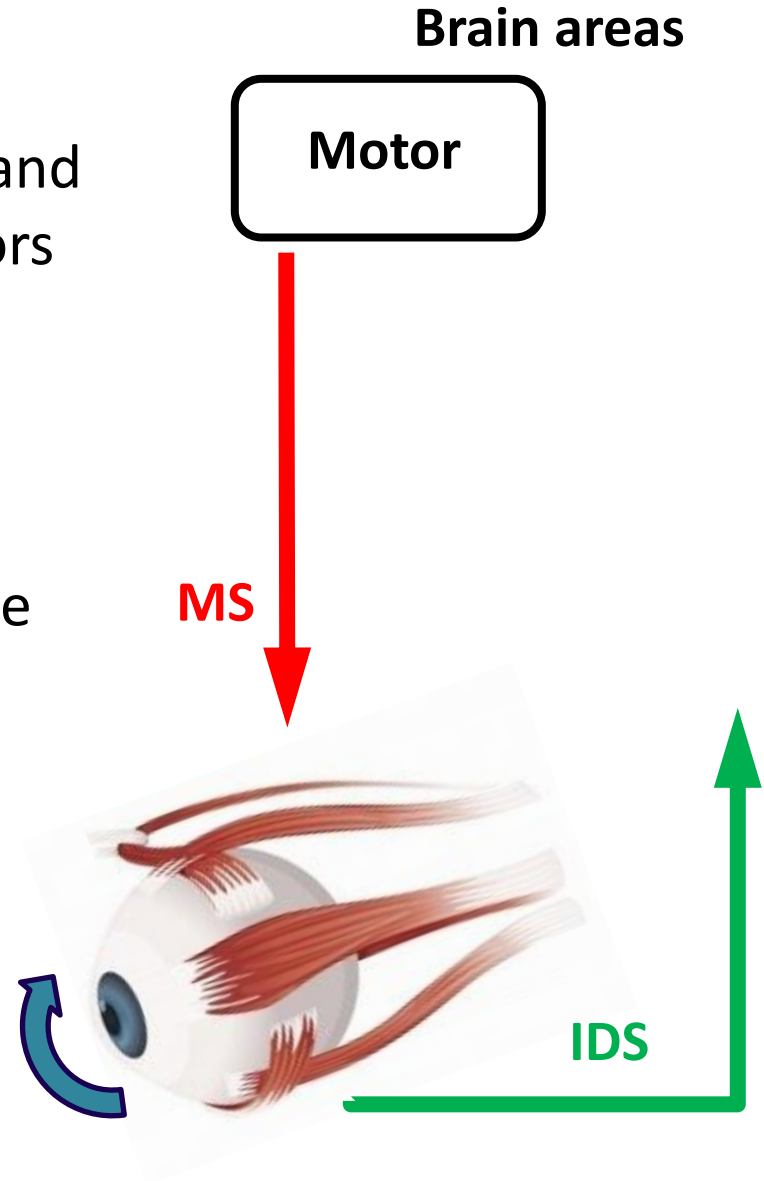
# Perception of real motion

## Image displacement signal (IDS)

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## Motor signal (MS)

A motion command is sent from the brain to the eye muscles.



# Perception of real motion

## Image displacement signal (IDS)

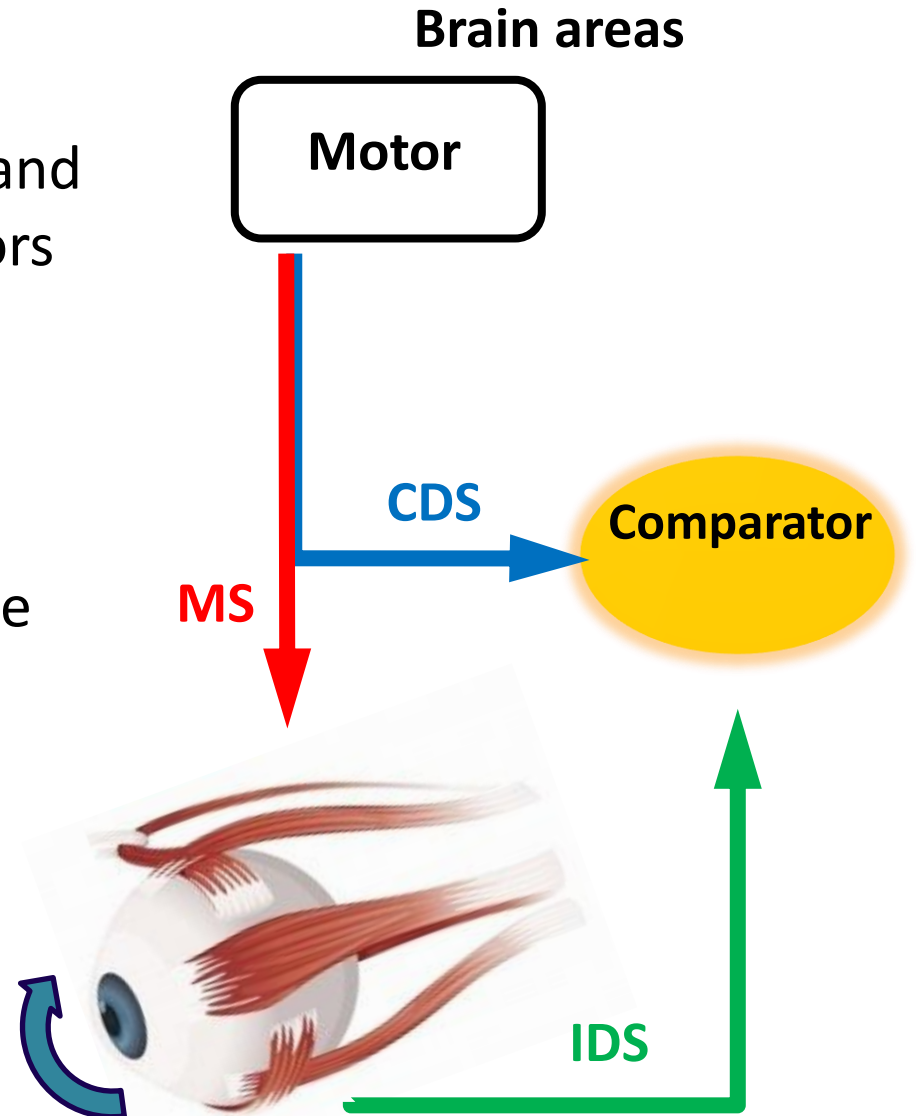
An image moves across the retina and successively stimulates the receptors (retinal motion).

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## Corollary discharge signal (CDS)

Copy of the motor signal ('Cc' to comparator).



# Perception of real motion

## Image displacement signal (IDS)

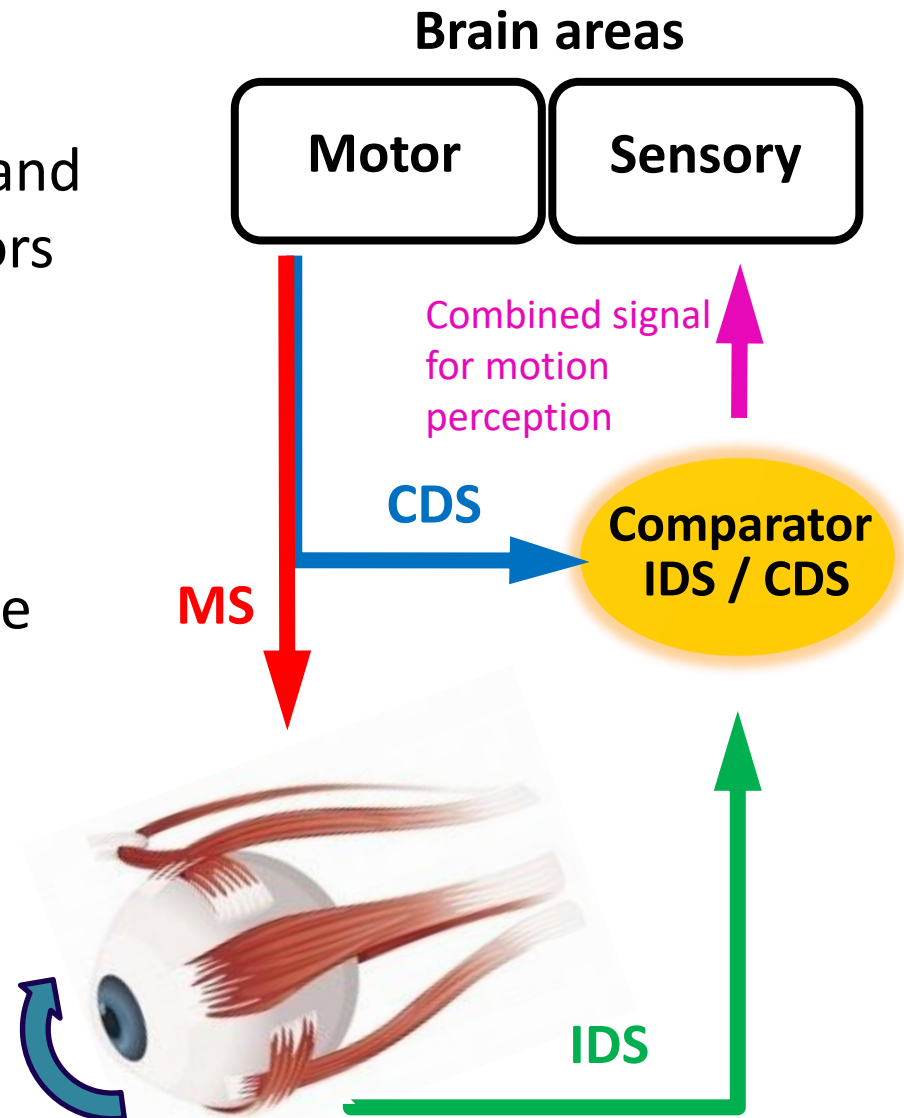
An image moves across the retina and successively stimulates the receptors (retinal motion).

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Copy of the motor signal ('Cc' to comparator).



# Perception of real motion

Is there...

...an image displacement signal? YES(1)

...a motor signal? YES(1)

...a corollary discharge signal? YES(1)

Static stimulus in a moving eye.



**B**



Brain areas

Motor

Sensory

No real motion

0

1  
CDS

Comparator  
IDS / CDS

1

IDS



# Perception of real motion

Moving stimulus



Static eye.

# Perception of real motion

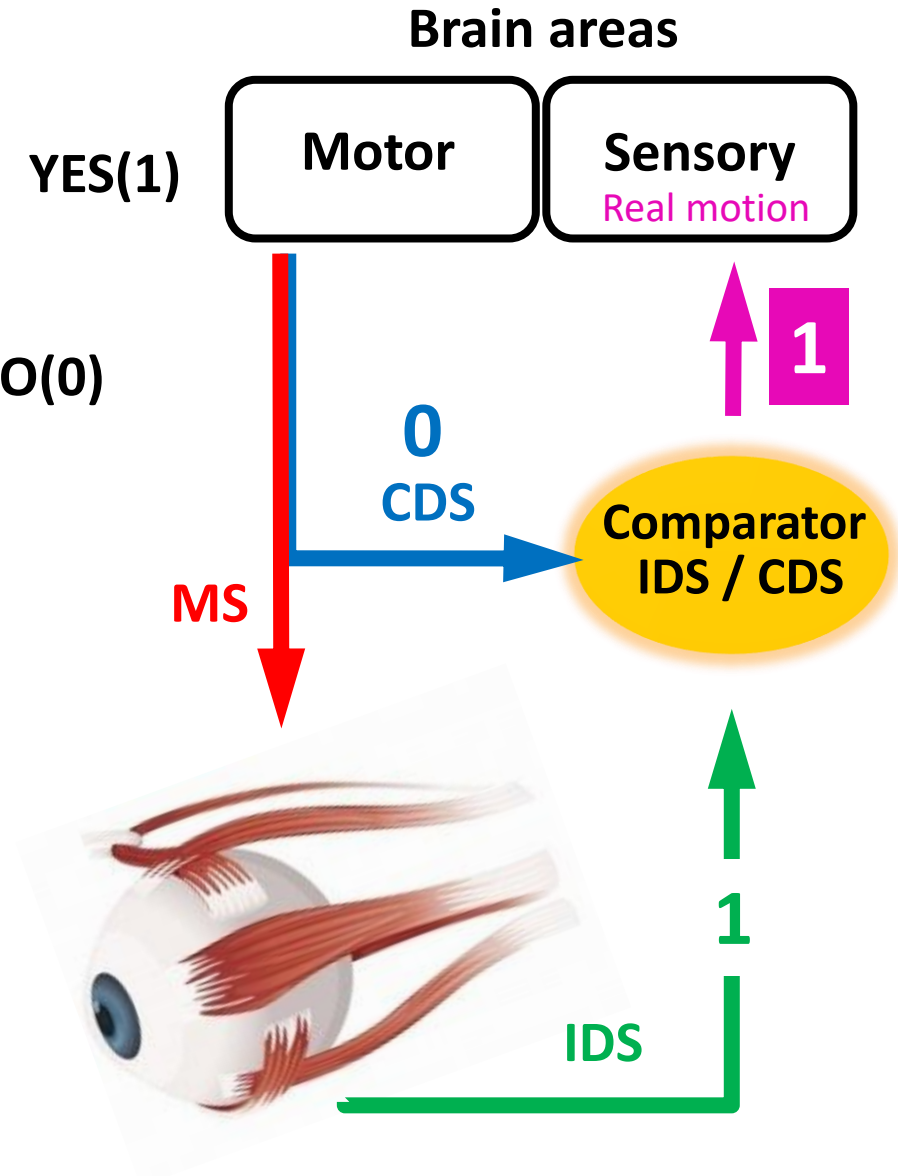
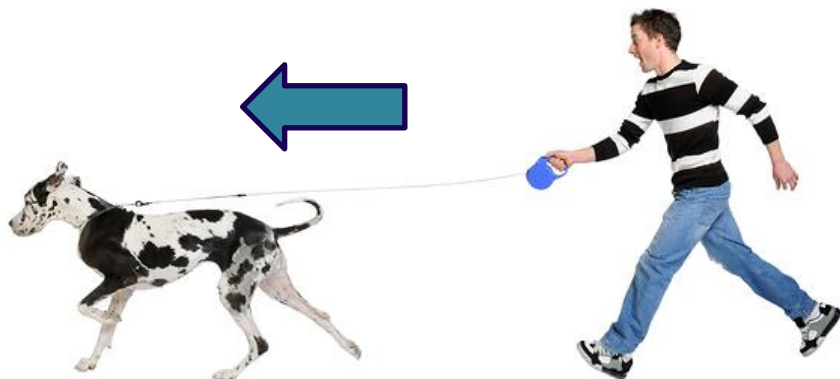
Is there...

...an image displacement signal? YES(1)

...a motor signal? NO(0)

...a corollary discharge signal? NO(0)

Moving stimulus in a static eye.



# Perception of real motion

Moving stimulus



Eye follows the moving stimulus.

# Perception of real motion

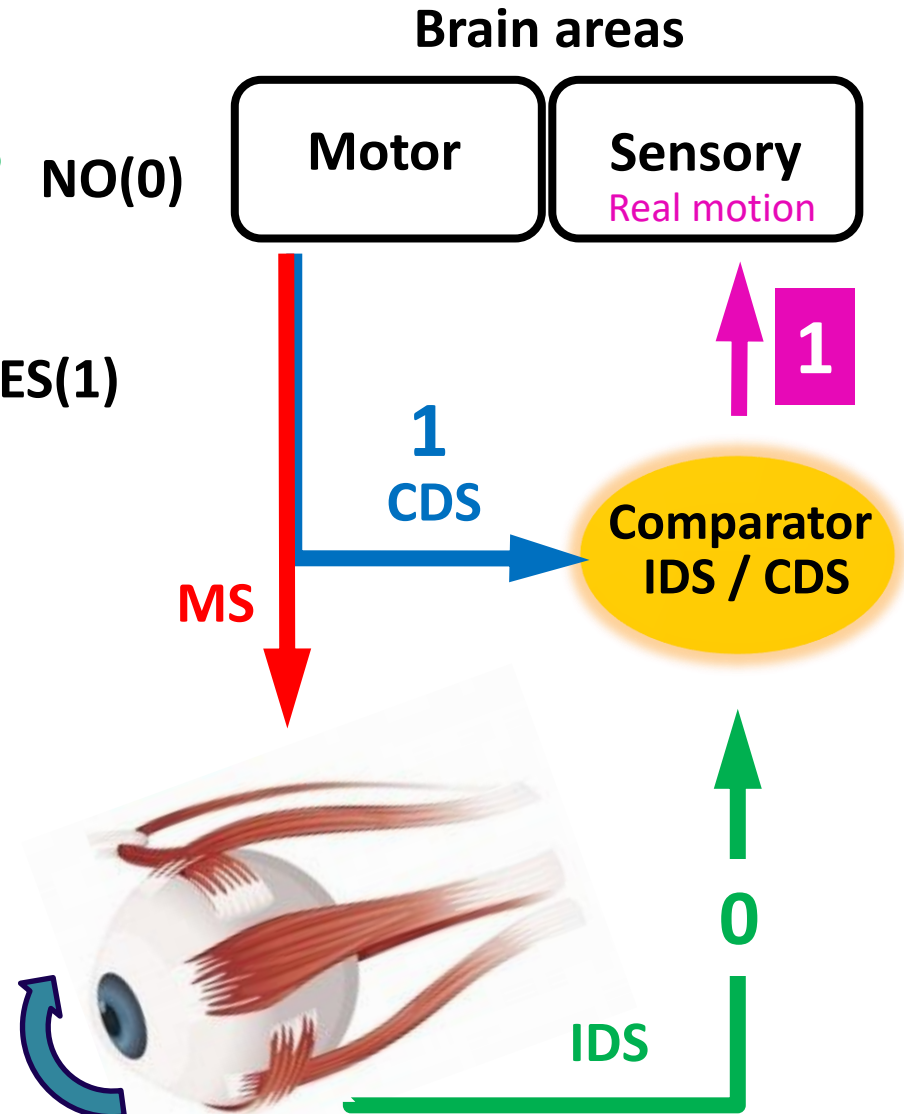
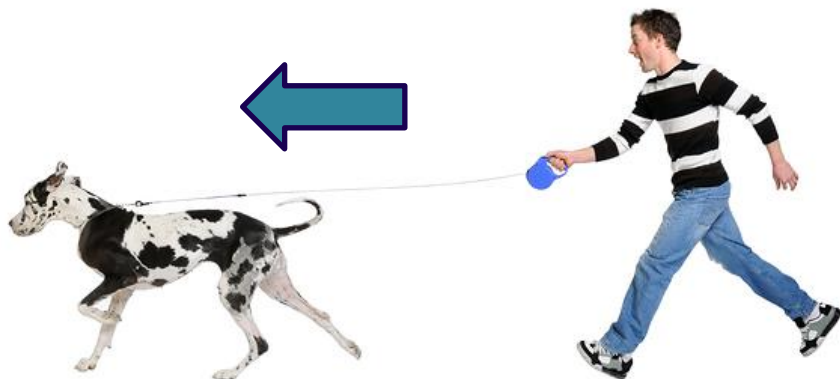
Is there...

...an image displacement signal? NO(0)

...a motor signal? YES(1)

...a corollary discharge signal? YES(1)

Moving stimulus in a moving eye.



# Perception of real motion

- Corollary Discharge Theory
  - Behavioural evidence
    - Motion perception matches the **model predictions**.

# Perception of real motion

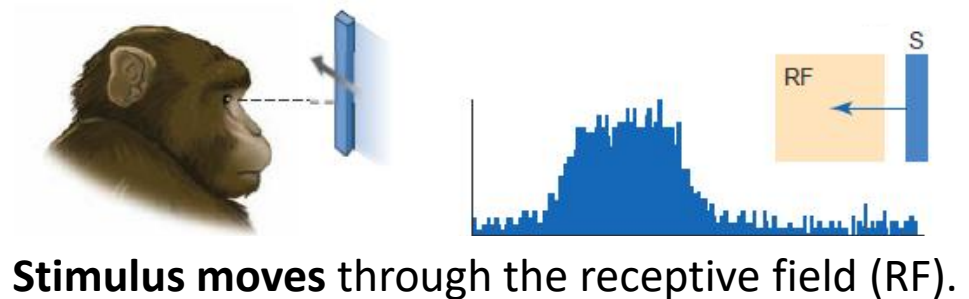
- Corollary Discharge Theory

- Neuronal evidence

Galletti & Fattori, 2003

**Real-motion neurons** respond to moving stimuli, but not to static stimuli in the moving eye.

Response of real-motion neurons (dorsal stream).





# Perception of real motion

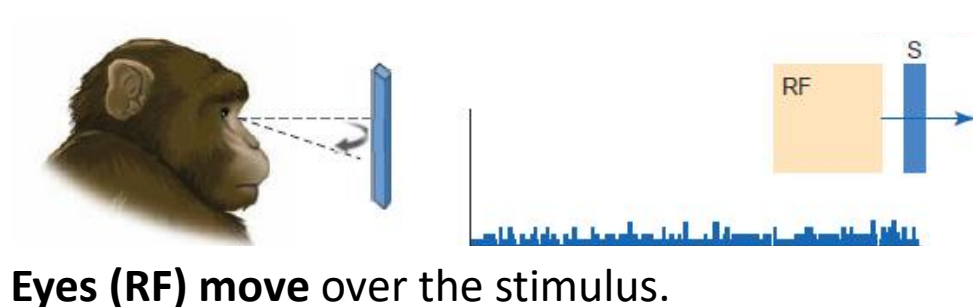
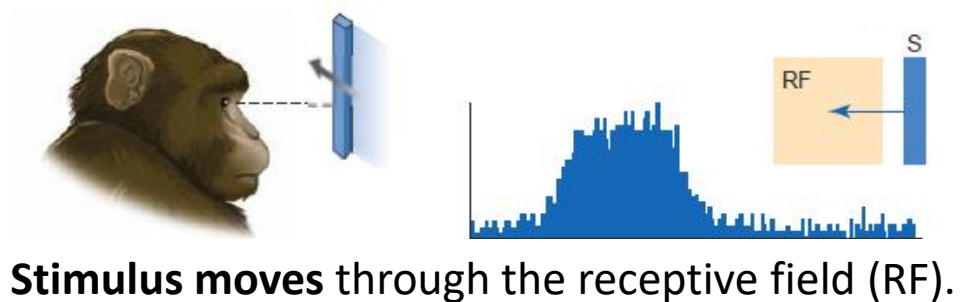
- Corollary Discharge Theory

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# Perception of real motion

- Corollary Discharge Theory

- Evaluation

Successfully explains why

a **moving stimulus** in a static or moving eye generates motion perception,

while a **static stimulus** in a moving eye does not, despite the fact that there is retinal motion.

# Summary

- We have looked at
  - **Real** versus **illusory motion** (e.g., apparent motion, induced motion, aftereffects, illusions).  
Only real motion produces **retinal motion**.
  - **Motion-direction selective neurons** (V1-V5) that process where an object is (evidence from single-cell recordings; Newsome et al., 1995).  
Neuronal response is pooled and there are end-stopped cells to solve the **aperture problem**.  
MT (dorsal) and IT (ventral) are interlinked in the superior temporal sulcus (**STS**), which makes this an expert area for perception and recognition of **biological motion**.
  - The **Corollary Discharge Theory** which explains motion perception by taking **retinal motion** (image displacement signal) **and eye movements** (motor signal, corollary discharge signal) into account (evidence from single-cell recordings; Galletti & Frattori, 2003).

# Test yourself – learning goals

- After reviewing this lecture you should be able to
  - Explain different ways we perceive motion and how they differ.
  - Define retinal motion and give examples for when it is produced and when not.
  - Name the most important brain areas involved in motion perception and explain why we know this.
  - Discuss the aperture problem and how the brain solves it.
  - Discuss the Corollary Discharge Theory, its assumptions, and the evidence for it.

## Test yourself – multiple choice

1. What do apparent motion and induced motion have in common?

- a) Both are biological motion.
- b) Both do not induce eye movements.
- c) Both do not produce retinal motion.
- d) Both need a reference frame.

## Test yourself – multiple choice

2. Which statement about a single neuron in the middle temporal area is not correct?

- a) It is most likely sensitive to motion direction.
- b) It may project to the superior temporal sulcus to support perception of biological motion.
- c) It is located in the dorsal visual pathway.
- d) It has a large receptive field covering the whole visual field.



## Test yourself – multiple choice

3. We are so good at detecting biological motion because:

- a) It produces retinal motion.
- b) It receives projects from both the dorsal and ventral visual pathway.
- c) It was invented by us (point-light walking).
- d) It is produced by us.

See you on Friday!

